Finding Bugs and Features Using Cryptographically-Informed Functional Testing

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Joint work with Giacomo Fenzi (EPFL Lausanne) and Jan Gilcher (ETH Zurich)

Real World Crypto 2026

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- Obviously, you may want to test your implementation

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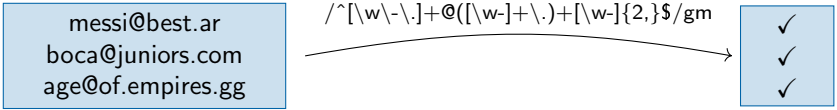

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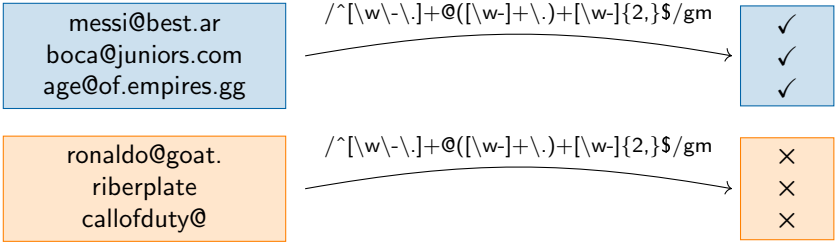

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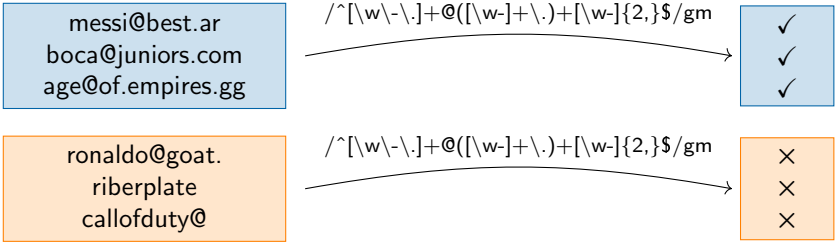

- However, we are not testing “predictable” functions

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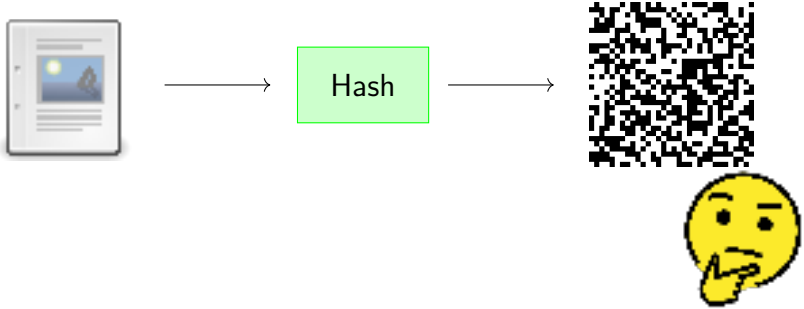

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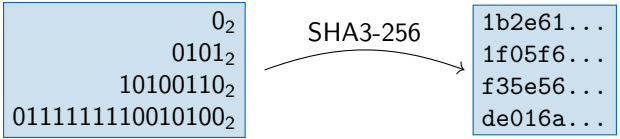

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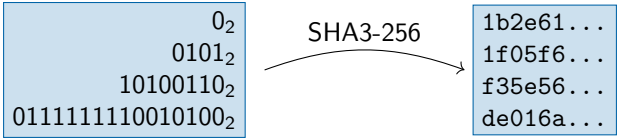

- You are checking against an existing implementation

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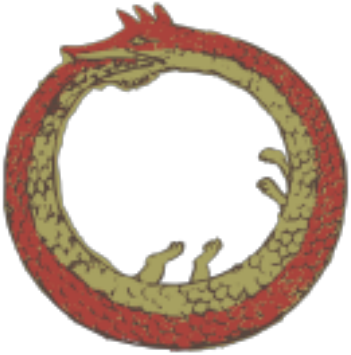

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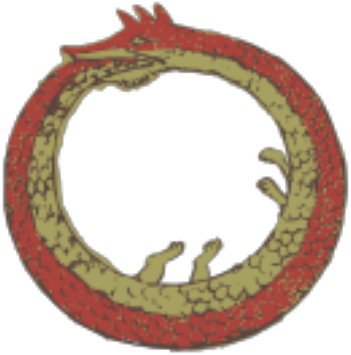

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- What if you were the designer of H? How do you test **before** KATs are available?

- Mouha *et al.* (IEEE Trans. on Reliability, 2018) observe: we don't know the output of H, but we know its cryptographic properties. — Use these properties to inform your testing!

Definition: collision resistance

Given $H : \{0,1\}^* \to \{0,1\}^n$, it's hard to find *distinct* $x, y \in \{0,1\}^*$ such that $H(x) = H(y)$.

Note: For hash functions is a cryptographic property *and a functional requirement*.

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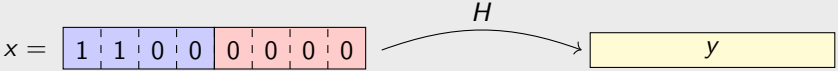

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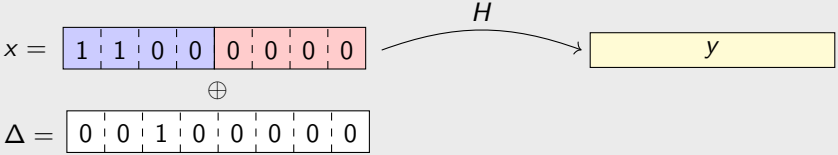

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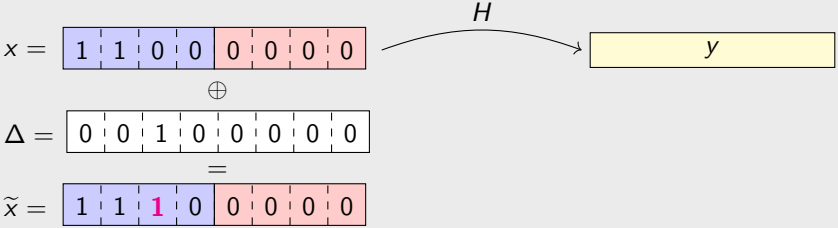

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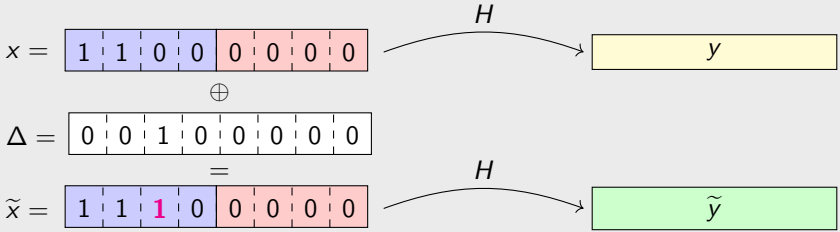

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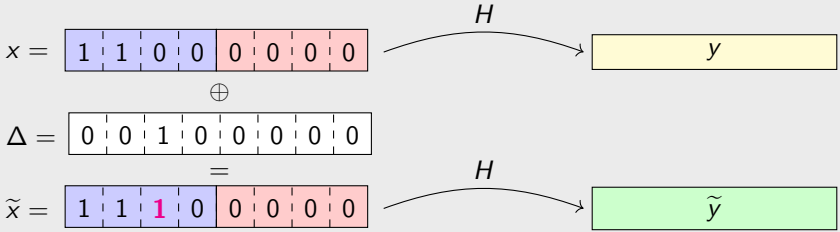

Check that $\tilde{y} \neq y$. Δ affects **output-contributing bits** of x, should result in different digests.

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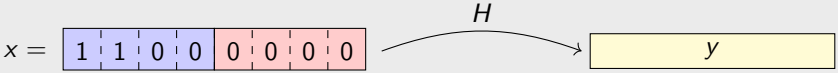

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- If $y \neq \widetilde{y}$, likely a buffer over-read!

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- In this talk: KEMs. For DSSs, see the paper.

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For a given $(\mathsf{sk}, \mathsf{pk})$ pair, challenge encapsulation c^* and key k^*, an adversary with access to a $\mathsf{Decaps}_{\mathsf{sk}}(\cdot)$ oracle cannot tell whether $\mathsf{Decaps}_{\mathsf{sk}}(\mathsf{c}^*) \stackrel{?}{=} \mathsf{k}^*$.

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- Informally, ciphertext is non-malleable!

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- Let's look at how we implemented this.

Decaps-Maul(c) Test (IND-CCA2 violations)

```
sus ← [ ];
Δ ← 1||0||...||0
(c, k, sk) ← GenInput()
repeat 8 · bytelen(c) times
    c̃, Δ′ ← Maul(c, Δ)
    try
        k̃ ← Decaps_sk(c̃)
        assert(k ≠ k̃)
    catch
        sus.append((c, Δ))
    Δ ← Δ′
return sus
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sus ← [ ];
Δ ← 1||0||...||0
(c, k, sk) ← GenInput()
repeat 8 · bytelen(c) times
    c̃, Δ′ ← Maul(c, Δ)
    try
        k̃ ← Decaps_sk(c̃)
        assert(k ≠ k̃)
    catch
        sus.append((c, Δ))
    Δ ← Δ′
return sus
```

GenInput()

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(sk, pk) ← Keygen()
(c, k) ← Encaps_pk()
return (c, k, sk)
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Maul(c, Δ)

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c̃ ← c ⊕ Δ
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Run in a fuzzer!

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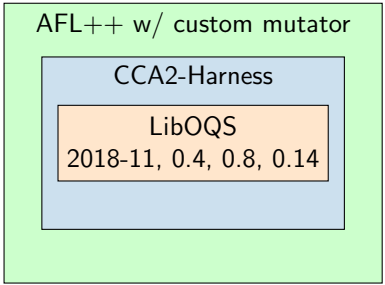

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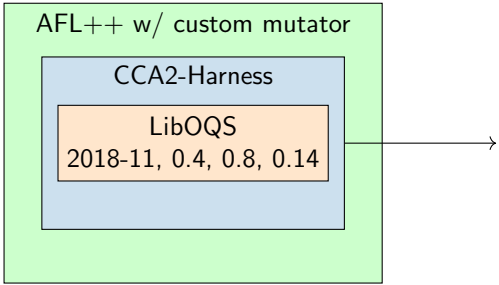

CCA2-Harness(sk, c, Δ, k)

1 : $\widetilde{k} \leftarrow \mathsf{Decaps}_{\mathsf{sk}}(\mathsf{c} \oplus \Delta)$

2 : **if** Δ contributes **and** $\widetilde{k} = k$

3 : `assert(0)`

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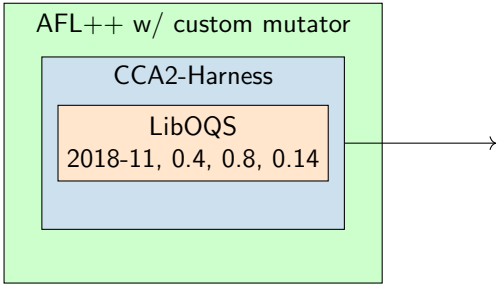

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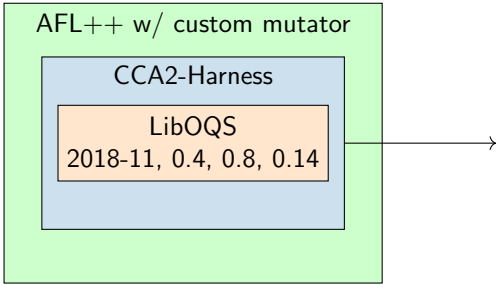

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- Before looking at results, let's run a quick poll.

Pop quiz

Consider an IND-CCA2 secure KEM with a **correct** implementation.

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Independent of γ

- There is no obvious negative consequence, but it is also not intuitive.

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- This is an artifact of testing so many schemes! For most individual schemes, testing is fast.

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- Against a "baseline" fuzzing campaign used as control, our tests only missed *few* SW bugs
- Cryptographically-informed testing is complementary to traditional fuzzing

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Thank you

Full paper: ia.cr/2024/1122, artefact: github.com/jangilcher/cryptoTesting